

SODERARM, Sweden

WMO: 024930

Lat: 59.753N

Lon: 19.405E

Elev: 14

StdP: 101.16

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.5	-7.3	-12.7	1.3	-8.1	-11.0	1.5	-6.6	19.5	-0.7	18.3	-1.8	8.5	60	0.860

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.0	22.6	19.7	21.3	18.7	20.1	17.6	20.5	22.1	19.1	20.8	18.0	19.6	5.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.6	21.7	18.4	13.3	20.3	17.2	12.3	19.0	59.3	22.3	54.5	20.8	50.8	19.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	16.0	14.3	DB	-10.1	23.8	4.2	1.6	-13.1	24.9	-15.6	25.8	-17.9	26.7	-21.0	27.8	(4)
(5)			WB	-10.5	20.3	2.5	1.3	-12.3	21.3	-13.8	22.0	-15.2	22.8	-17.0	23.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-0.5	-1.1	0.4	3.6	7.9	12.8	17.1	17.2	13.2	8.3	4.5	1.7
	DBStd	6.93	3.23	3.43	2.81	2.03	2.27	2.41	2.40	2.11	2.13	2.64	2.67	3.25
	HDD10.0	1692	326	310	297	193	72	4	0	0	3	64	165	257
	HDD18.3	4111	584	543	555	443	324	165	55	47	154	310	415	516
	CDD10.0	649	0	0	0	0	7	89	219	224	99	12	0	0
	CDD18.3	28	0	0	0	0	0	0	16	12	0	0	0	0
	CDH23.3	16	0	0	0	0	0	0	12	4	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.6	9.1	8.2	7.7	6.9	6.5	6.3	6.1	6.3	7.3	8.3	8.9	9.3
Precipitation	PrecAvg	495	37	29	24	27	34	48	46	57	49	49	50	45
	PrecMax	721	79	54	52	65	75	95	116	141	116	110	106	94
	PrecMin	366	2	4	1	4	7	16	5	1	6	1	14	10
	PrecStd	76	19	12	13	17	17	21	29	34	28	23	23	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	5.6	7.2	10.5	15.2	19.7	24.7	23.7	18.7	14.6	10.1	8.7
		MCWB	4.8	3.6	4.8	7.1	11.1	15.6	20.4	21.0	16.3	13.4	9.1	7.2
	2%	DB	5.0	4.4	5.7	8.6	13.3	18.4	23.1	22.3	17.6	13.3	9.3	7.2
		MCWB	4.1	3.0	3.8	6.0	10.4	15.0	20.0	19.5	15.4	11.9	8.4	6.1
	5%	DB	4.3	3.6	4.6	7.4	12.0	17.2	21.9	21.2	16.7	12.5	8.5	6.3
		MCWB	3.5	2.5	3.1	5.4	9.7	14.6	19.4	18.6	14.8	11.3	7.7	5.4
	10%	DB	3.5	2.8	3.7	6.3	10.9	16.3	20.7	20.1	16.0	11.8	7.8	5.6
		MCWB	2.6	1.9	2.5	4.7	9.1	14.1	18.3	17.6	14.3	10.7	7.1	4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	4.2	5.1	7.5	12.1	16.8	22.1	22.2	17.0	13.8	9.6	7.2
		MCDB	5.9	4.9	6.7	9.8	13.9	18.5	23.0	23.2	17.9	14.2	9.8	7.6
	2%	WB	4.4	3.3	4.1	6.4	10.9	15.9	20.9	20.5	16.0	12.3	8.6	6.1
		MCDB	4.9	4.1	5.4	8.1	12.6	17.3	22.6	21.8	17.1	12.9	9.1	6.5
	5%	WB	3.6	2.7	3.3	5.6	10.2	15.1	19.7	19.1	15.2	11.6	8.0	5.3
		MCDB	4.2	3.4	4.3	7.0	11.7	16.7	21.4	20.6	16.4	12.3	8.4	5.8
	10%	WB	2.8	2.1	2.7	4.9	9.4	14.4	18.4	18.1	14.7	10.9	7.2	4.6
		MCDB	3.5	2.8	3.5	6.1	10.7	16.0	20.2	19.7	15.8	11.6	7.7	5.3
Mean Daily Temperature Range	5% DB	MDBR	2.8	2.8	2.9	3.3	3.5	3.6	3.3	3.0	2.7	2.4	2.4	2.8
		MCDBR	2.9	2.9	3.9	5.1	5.2	5.0	4.4	3.6	3.1	2.8	2.5	2.9
		MCWBR	2.8	2.7	2.8	3.1	3.2	2.9	2.6	2.5	2.3	2.5	2.6	2.6
	5% WB	MCDBR	2.8	2.8	3.6	4.8	4.7	4.4	3.9	3.2	2.8	2.7	2.2	2.7
		MCWBR	2.8	2.7	2.8	3.1	3.2	2.8	2.7	2.6	2.3	2.7	2.3	2.7
Clear-Sky Solar Irradiance	taub	0.282	0.317	0.335	0.373	0.369	0.361	0.389	0.379	0.348	0.335	0.307	0.265	
	taud	2.210	2.284	2.340	2.335	2.421	2.487	2.427	2.443	2.513	2.416	2.237	2.120	
	Ebn at Noon	501	684	797	827	860	873	835	814	785	662	457	362	
	Edh at Noon	42	68	87	104	103	98	102	93	74	60	41	31	
All-Sky Solar Radiation	RadAvg	0.26	0.80	2.31	4.01	5.40	6.01	5.61	4.39	2.77	1.19	0.36	0.17	
	RadStd	0.03	0.15	0.24	0.36	0.44	0.35	0.52	0.36	0.29	0.19	0.08	0.02	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-110	-131	N/A	N/A

Nomenclature: See separate page